

Seanie Lee

DATE OF BIRTH: 1992.04.17, NATIONALITY: KOREAN

Seattle, Washington

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Education

KAIST (Korea Advanced Institute of Science and Technology)

Daejeon, S.Korea

PH.D IN ARTIFICIAL INTELLIGENCE

2022 - 2026

- Supervised by [Sung Ju Hwang](#) and [Juho Lee](#)
- Research interest: AI safety, responsible AI, and evaluation

KAIST (Korea Advanced Institute of Science and Technology)

Daejeon, S.Korea

M.S. IN ARTIFICIAL INTELLIGENCE

2020 - 2022

- Supervised by [Sung Ju Hwang](#) and [Juho Lee](#)
- Master Thesis: [Data augmentation for natural language processing](#)

Yonsei University

Seoul, S.Korea

B.A. IN LIBRARY AND INFORMATION SCIENCE

2011 - 2018

Research Interests

- Data-Centric AI [[2](#), [4](#), [5](#), [9](#), [16](#), [19](#), [20](#), [25](#), [27](#), [31](#), [32](#)]
- Responsible and Safe AI [[19](#), [20](#), [22](#), [23](#), [24](#), [26](#), [27](#), [29](#), [30](#), [31](#)]
- Efficient Machine Learning [[7](#), [13](#), [16](#), [18](#), [20](#), [22](#), [25](#)]
- Agentic AI [[25](#), [29](#), [30](#), [32](#)]

Experience

Apple

Seattle, US

INTERNSHIP

October 2025 - September 2026

- Machine Learning Research, hosted by [Raviteja Vemulapalli](#).
- Synthetic data generation for tool-calling LLMs.

Krafton

Seoul, Korea

INTERNSHIP

July 2025 - Oct 2025

- Research Internship
- Safety alignment of long reasoning models.

Mila

Montreal, Canada

INTERNSHIP

January 2024 - June 2024

- Research internship at Mila, advised by [Yoshua Bengio](#).
- Robust red-teaming of LLMs.

Apple

Cambridge, UK

INTERNSHIP

May 2023 - September 2023

- Research internship at Siri team, hosted by [Anders Johannsen](#).
- Few-shot example retrieval for ICL.

National University of Singapore (NUS)

Singapore

INTERNSHIP

July 2022 - September 2022

- Remote internship at [Deep Learning](#) lab, supervised by [Kenji Kawaguchi](#).
- Regularization for continual pretraining.

Awards

2023 **Apple AI/ML PhD Fellowship**, Recipient of [Apple Scholars in AI/ML](#)

Cupertino, US

2022 **Google Travel Grant**, NeurIPS 2022

US

2019 **Silver Medal**, Named Entity Recognition in [NAVER NLP Challenge](#)

Seoul, Korea

Presentation

Seminar at Korea University. PRESENTATION OF LARGE SCALE SET-ENCODING - Synthetic Data Generation for LLM Safeguards - ICLR 2025, ACL Findings 2025	Seoul, Korea May. 2025
Seminar at Hanyang University. PRESENTATION OF LARGE SCALE SET-ENCODING - Synthetic Data Generation for LLM Safeguards - ICLR 2025, ACL Findings 2025	Seoul, Korea April. 2025
Tech. Talk, Nuremberg Institute of Technology Georg Simon Ohm. PRESENTATION OF LARGE SCALE SET-ENCODING - Scalable Set Encoding with Universal Mini-Batch Consistency and Unbiased Full Set Gradient Approximation - ICML 2023	Nürnberg, Germany Oct. 2023
Tech. talk, Samsung SDS. PRESENTATION OF LARGE SCALE SET-ENCODING - Scalable Set Encoding with Universal Mini-Batch Consistency and Unbiased Full Set Gradient Approximation - ICML 2023	Seoul, South Korea 22.May. 2023
Tech. talk, NAVER corp. PRESENTATION OF INFO-HCVAE - Generating Diverse and Consistent QA pairs from Contexts with Information-Maximizing Hierarchical Conditional VAEs - ACL 2020 Long paper	Online, South Korea 04.Dec. 2020

Publications

(* indicates equal contribution)

PREPRINTS

[32] Simulate to Generalize: Scaling Stateful Supervision for API-calling Agents using LLM World Models SEANIE LEE, SANJOY CHOWDHURY, CHAO JIANG, CHENG-YU HSIEH, TING-YAO HU, ALEXANDER T TOSHEV, ONCEL TUZEL AND RAVITEJA VEMULAPALLI [paper]	arXiv 2026
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CONFERENCES AND JOURNALS

[31] THINKSAFE: Self-Generated Safety Alignment for Reasoning Models SEANIE LEE*, SANGWOO PARK*, YUMIN CHOI, GYEONGMAN KIM, MINKI KANG, JIHUN YUN, DONGMIN PARK, JONGHO PARK AND SUNG JU HWANG [paper][code]	NeurIPS 2026
[30] It Takes Two: Complementary Self-Distillation for Contextual Integrity in LLMs SANGWOO PARK*, WOONGYEONG YEO*, SEANIE LEE, YUMIN CHOI, HYOMIN LEE, KANGSAN KIM, JINHEON BAEK, SEONG JOON OH AND SUNG JU HWANG [paper][code]	NeurIPS 2026
[29] T-MAP: Red-Teaming LLM Agents with Trajectory-aware Evolutionary Search HYOMIN LEE, SANGWOO PARK, YUMIN CHOI, SOHYUN AN, SEANIE LEE AND SUNG JU HWANG [paper][code]	EMNLP 2026
[28] Rethinking Reward Models for Multi-Domain Test-Time Scaling DONG BOK LEE*, SEANIE LEE*, SANGWOO PARK, MINKI KANG, JINHEON BAEK, DONGKI KIM, DOMINIK WAGNER, JIONGDAO JIN, HEEJUN LEE, TOBIAS BOCKLET, JINYU WANG, JINGJING FU, SUNG JU HWANG, JIANG BIAN AND LEI SONG [paper][code]	TMLR 2026
[27] HoliSafe: Holistic Safety Benchmarking and Modeling with Safety Meta Token for Vision-Language Model YOUNGWAN LEE, KANGSAN KIM, KWANYONG PARK, ILCAHE JUNG, SOOJIN JANG, SEANIE LEE, YONG-JU LEE AND SUNG JU HWANG [paper][code]	CVPR Finding 2025

[26] FedSVD: Adaptive Orthogonalization for Private Federated Learning with LoRA SEANIE LEE*, SANGWOO PARK*, DONG BOK LEE*, DOMINIK WAGNER, HAEBIN SEONG, TOBIAS BOCKLET, JUHO LEE, SUNG JU HWANG • [paper][code]	<i>NeurIPS</i> 2025
[25] Distilling LLM Agent into Small Models with Retrieval and Code Tools MINKI KANG, JONGWON JEONG, SEANIE LEE, JAEWOONG CHO AND SUNG JU HWANG • [paper][code]	<i>NeurIPS Spotlight</i> 2025
[24] Reliable Decision-Making via Calibration-Oriented Retrieval-Augmented Generation CHAEYUN JANG, DEUKHWAN CHO, SEANIE LEE, HYUNGI LEE AND JUHO LEE • [paper][code]	<i>NeurIPS</i> 2025
[23] Trajectory Balance with Asynchrony: Decoupling Exploration and Learning for Fast, Scalable LLM Post-Training BRIAN R. BARTOLDSON, SIDDARTH VENKATRAMAN, JAMES DIFFENDERFER, MOKSH JAIN, TAL BEN-NUN, SEANIE LEE, MINSU KIM, JOHAN OBANDO-CERON, YOSHUA BENGIO AND BHAVYA KAILKHURA • [paper][code]	<i>NeurIPS</i> 2025
[22] SafeRoute: Adaptive Model Selection for Efficient and Accurate Safety Guardrails in Large Language Models SEANIE LEE*, DONG BOK LEE*, DOMINIK WAGNER, MINKI KANG, HAEBIN SEONG, TOBIAS BOCKLET, JUHO LEE, SUNG JU HWANG • [paper][code]	<i>ACL Findings</i> 2025
[21] Personalized Fine-Tuning with Controllable Synthetic Speech from LLM-Generated Transcripts for Dysarthric Speech Recognition DOMINIK WAGNER, ILJA BAUMANN, NATALIE ENGERT, SEANIE LEE, ELMAR NÖTH, KORBINIAN RIEDHAMMER AND TOBIAS BOCKLET • [paper]	<i>Interspeech</i> 2025
[20] HarmAug: Effective Data Augmentation for Knowledge Distillation of Safety Guard Models SEANIE LEE*, HAEBIN SEONG*, DONG BOK LEE, MINKI KANG, XIAOYIN CHEN, DOMINIK WAGNER, YOSHUA BENGIO, JUHO LEE, SUNG JU HWANG • [paper][code]	<i>ICLR</i> 2025
[19] Learning Diverse Attacks on Large Language Models for Robust Red-teaming and Safety Tuning SEANIE LEE, MINSU KIM, LYNN CHERIF, DAVID DOBRE, JUHO LEE, SUNG JU HWANG, KENJI KAWAGUCHI, GAUTHIER GIDEL, YOSHUA BENGIO, NIKOLAY MALKIN, MOKSH JAIN • [paper][code]	<i>ICLR</i> 2025
[18] Optimized Speculative Sampling for GPU Hardware Accelerators DOMINIK WAGNER, SEANIE LEE, ILJA BAUMANN, PHILIPP SEEBERGER, KORBINIAN RIEDHAMMER, TOBIAS BOCKLET • [paper][code]	<i>EMNLP</i> 2024
[17] Drug Discovery with Dynamic Goal-aware Fragment SEUL LEE, SEANIE LEE, KENJI KAWAGUCHI, SUNG JU HWANG • [paper][code]	<i>ICML</i> 2024
[16] Effective and Efficient Conversation Retrieval for Dialogue State Tracking with Implicit Text Summaries SEANIE LEE, JIANPENG CHENG, JORIS DRIESEN, ALEXANDRU COCA, ANDERS JOHANNSEN • [paper]	<i>NAACL</i> 2024
[15] Self-Supervised Dataset Distillation for Transfer Learning DONG BOK LEE*, SEANIE LEE*, JOONHO KO, KENJI KAWAGUCHI, JUHO LEE, SUNG JU HWANG • [paper][code]	<i>ICLR</i> 2024
[14] DiffusionNAG: Task-guided Neural Architecture Generation with Diffusion Models SOHYUN AHN*, HAYEON LEE*, JAEHYEONG JO, SEANIE LEE, SUNG JU HWANG • [paper][code]	<i>ICLR</i> 2024

[13] Scalable Set Encoding with Universal Mini-Batch Consistency and Unbiased Full Set Gradient Approximation JEFFREY WILLETTE*, SEANIE LEE* , BRUNO ANDREIS, KENJI KAWAGUCHI, JUHO LEE, SUNG JU HWANG • [paper] [code]	ICML 2023
[12] Margin-based Neural Network Watermarking BYUNGJOO KIM, SUYOUNG LEE, SEANIE LEE , SOOEL SON, SUNG JU HWANG • [paper]	ICML 2023
[11] Self-Supervised Set Representation Learning for Unsupervised Meta-Learning DONG BOK LEE*, SEANIE LEE* , KENJI KAWAGUCHI, YUNJI KIM, JIHWAN BANG, JUNG-WOO HA, SUNG JU HWANG • [paper]	ICLR 2023
[10] Self-Distillation for Further Pre-training of Transformers SEANIE LEE , MINKI KANG, JUHO LEE, SUNG JU HWANG, KENJI KAWAGUCHI • [paper] [code]	ICLR 2023
[9] Set-based Meta-Interpolation for Few-Task Meta-Learning SEANIE LEE* , BRUNO ANDREIS*, KENJI KAWAGUCHI, SUNG JU HWANG • [paper] [code]	NeurIPS 2022
[8] On Divergence Measures for Bayesian Pseudocoresets BALHAE KIM, JUNGWON CHOI, SEANIE LEE , YOONHO LEE, JUNG-WOO HA, JUHO LEE • [paper]	NeurIPS 2022
[7] Set Based Stochastic Subsampling BRUNO ANDREIS, SEANIE LEE , A. TUAN NGUYEN, JUHO LEE, EUNHO YANG, SUNG JU HWANG • [paper]	ICML 2022
[6] Sequential Reptile: Inter-Task Gradient Alignment for Multilingual Learning SEANIE LEE* , HAE BEOM LEE*, JUHO LEE, SUNG JU HWANG • [paper]	ICLR 2022
[5] Learning to Perturb Word Embeddings for Out-of-distribution QA SEANIE LEE* , MINKI KANG*, JUHO LEE, SUNG JU HWANG • [paper] [code]	ACL 2021
[4] Contrastive Learning with Adversarial Perturbations for Conditional Text Generation SEANIE LEE* , DONG BOK LEE*, SUNG JU HWANG • [paper] [code]	ICLR 2021
[3] Meta-GMVAE: Mixture of Gaussian VAE for Unsupervised Meta-Learning DONG BOK LEE, DONGCHAN MIN, SEANIE LEE , SUNG JU HWANG • [paper] [code]	ICLR 2021
[2] Generating Diverse and Consistent QA pairs from Contexts with Information-Maximizing Hierarchical Conditional VAEs DONG BOK LEE*, SEANIE LEE* , WOOTAE JEONG, DONGHWAN KIM, SUNG JU HWANG • [paper] [code] [video]	ACL 2020
[1] g2pM: A Neural Grapheme-to-Phoneme Conversion Package for Mandarin Chinese Based on a New Open Benchmark Dataset KYUBYONG PARK*, SEANIE LEE* • [paper] [code]	Interspeech 2020

References

Sung Ju Hwang ASSOCIATE PROFESSOR IN KAIST. e-mail: sjhwang82@kaist.ac.kr.	Advisor 2020-2025
Juho Lee ASSOCIATE PROFESSOR IN KAIST. e-mail: juholee@kaist.ac.kr.	Advisor 2020-2025

Yoshua Bengio

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Collaborator
2024-2025

Kenji Kawaguchi

PRESIDENTIAL YOUNG PROFESSOR IN THE DEPARTMENT OF COMPUTER SCIENCE AT NUS.
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Collaborator
2022-present

Nikolay Malkin

CHANCELLOR’S FELLOW AT UNIVERSITY OF EDINBURGH, SCHOOL OF INFORMATICS
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Collaborator
2024